

# Supplementary Information

## Storm-tide magnitude and standardized connected terrain yield non-interchangeable coastal screening priorities in a Northwest European station network

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This Supplementary Information is generated from the same frozen source tables as the manuscript. Legacy degree-window, 15 m denominator and category-based tables are intentionally excluded.

### Supplementary Methods

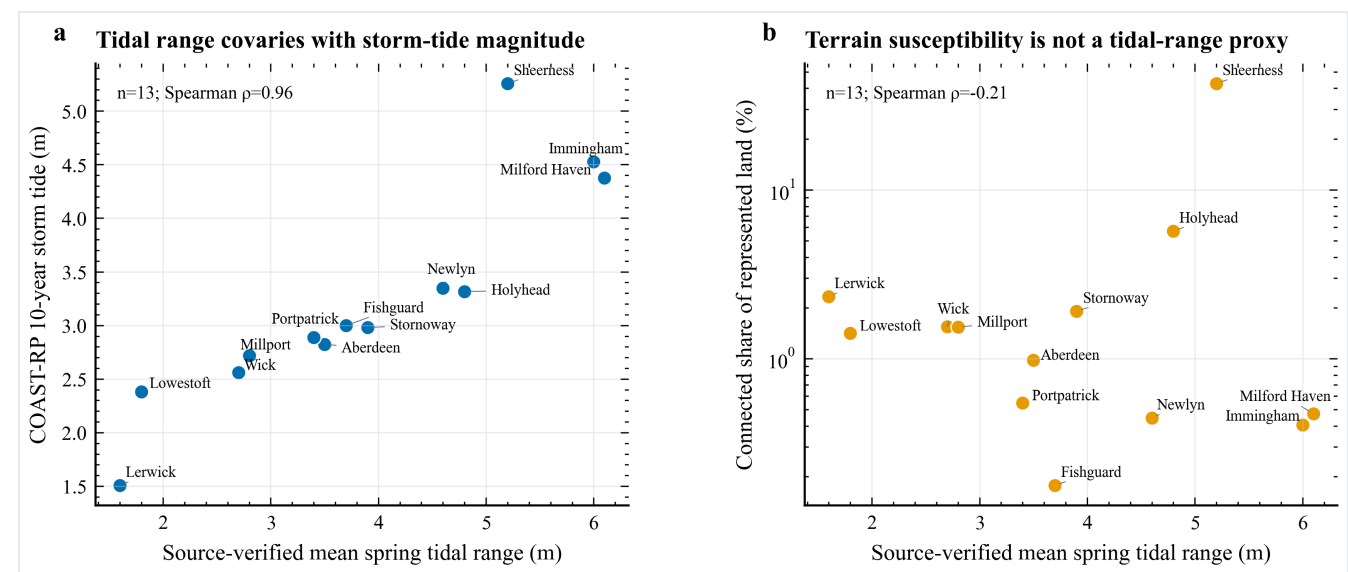
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The complete terrain grid contains 74 stations and 40 configurations per station (2,960 rows). The primary configuration is a 10 km window, standardized 2 m EGM2008 terrain threshold, four-neighbour graph, adaptive official-mask marine seed and COAST-RP match within 6 km. Only finite official-mask class-0 terrain enters the reference-land denominator. Internal outside-support value 254 and official classes 1, 2, 3 and 255 are excluded. Sector resampling, within-sector permutation and sector adjustment use the same six geographic identifiers shown in Fig. 1.

The local terrain comparison retains each source product's native vertical datum. Continuous elevation was downscaled by the provider WCS; the service interpolation kernel was not controlled by the study code. Categorical masks and the connected mask used for intersection-over-union were reprojected by nearest neighbour to the local WCS grid.

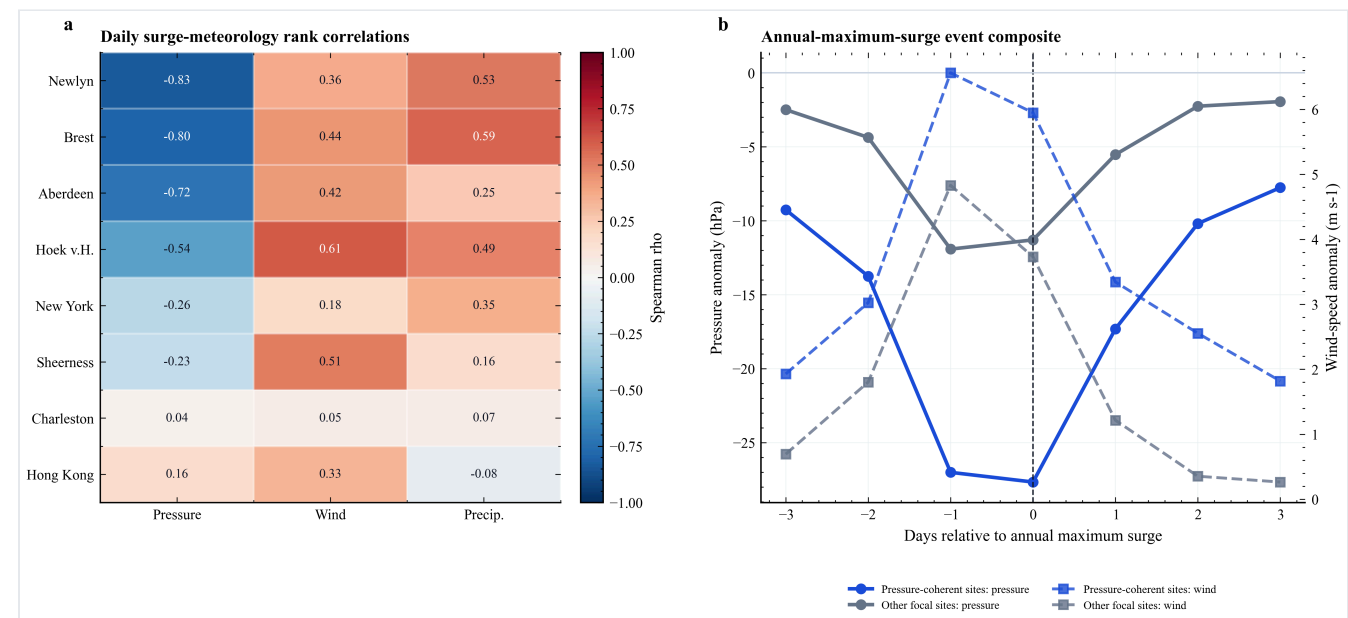
# Supplementary Figures

Supplementary Figure S1. Source-verified tidal context



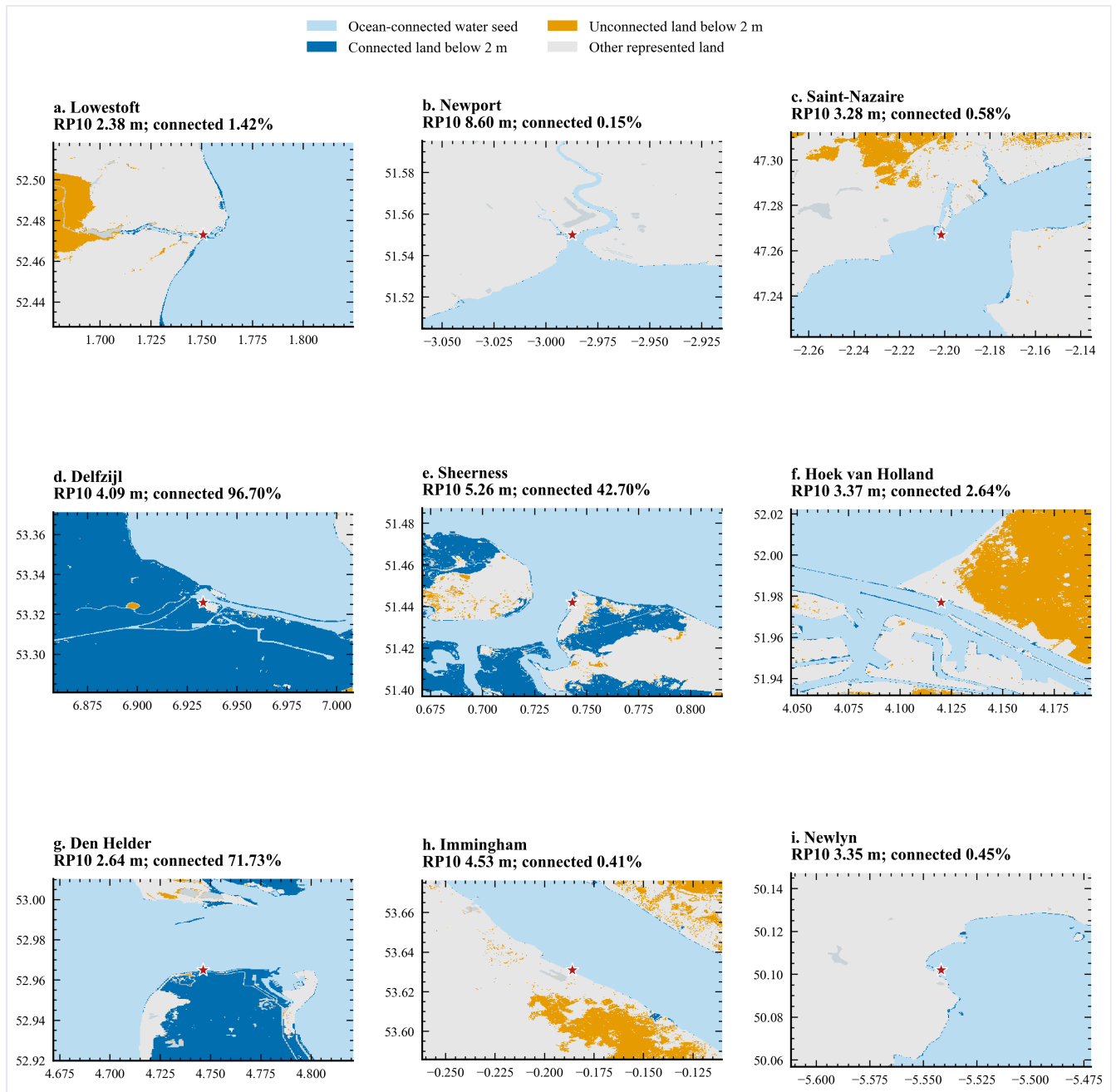
All 13 source-verified stations are plotted once. Tidal range covaries strongly with storm-tide magnitude but not with the standardized connected-terrain share in this small exploratory subset.

Supplementary Figure S2. Predictor-consistency diagnostic for focal GSSR series



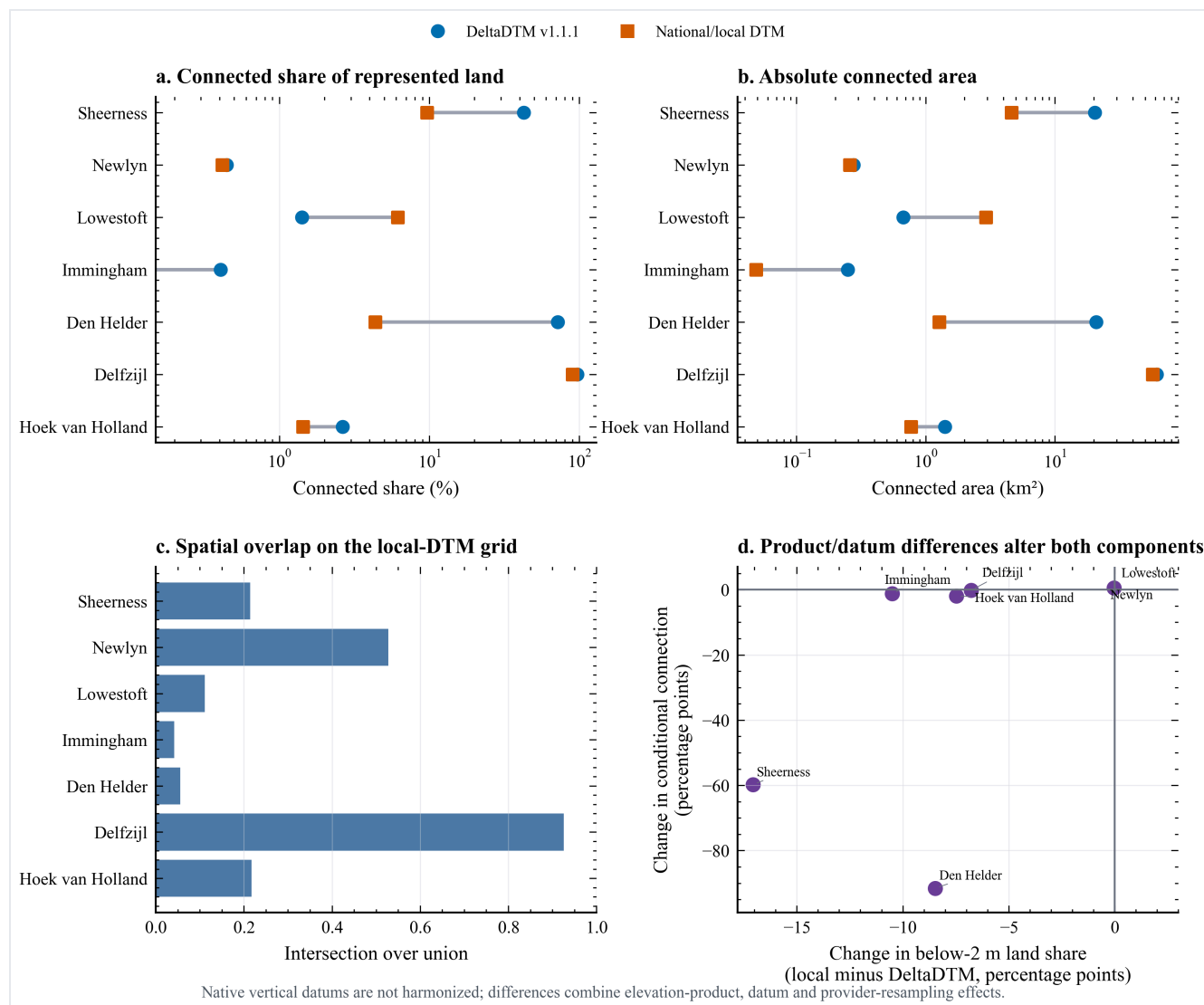
The atmospheric predictors overlap with the reconstruction framework. This panel is therefore not independent validation or extreme-event skill assessment and is not used in the terrain analysis.

## Supplementary Figure S3. Standardized static terrain gallery



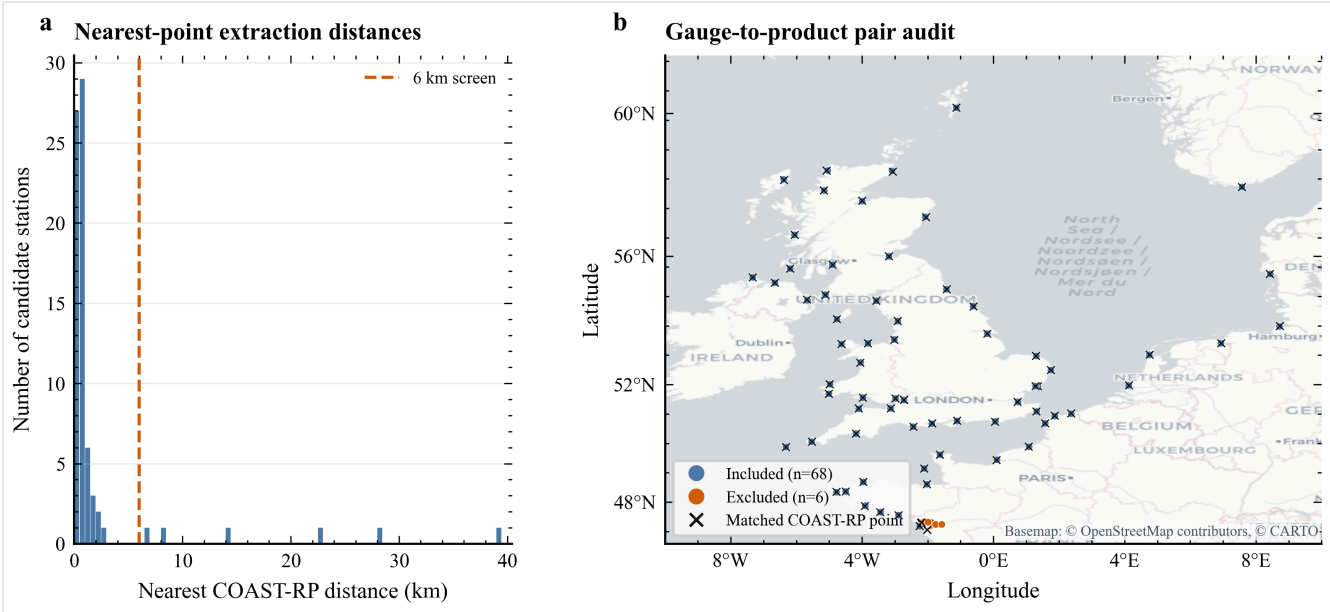
Every panel uses the same fixed 10 km window, official mask and 2 m EGM2008 threshold. Blue and orange cells retain their original raster footprint.

Supplementary Figure S4. Local-DTM product, datum and resampling sensitivity



The three vertical references are not harmonized. The figure combines elevation-product, native-datum and provider-resampling effects and is not a validation result.

Supplementary Figure S5. COAST-RP nearest-point matching audit



All 74 candidate gauge-to-product pairs are retained in the source table. The six excluded Loire estuary stations lie 6.90-39.42 km from the nearest COAST-RP point.

## Supplementary Tables

Supplementary Table S1. Frozen sample flow.

| stage                                                                  | n  |
|------------------------------------------------------------------------|----|
| GSSR metadata stations inside geographic bounds                        | 74 |
| DeltaDTM v1.1 elevation tiles available                                | 74 |
| Official marine seed resolved in 10 km window/context                  | 74 |
| COAST-RP nearest match within 6 km (primary)                           | 68 |
| GSSR record $\geq 25$ years and reconstruction correlation $\geq 0.55$ | 44 |

Supplementary Table S2. DeltaDTM v1.1.1 product-version and mask-fill audit.

| product  | release | tile tag                             | vertical reference | release cap m | paper v1 0 cap m | mask outside support fill | official 255 cells in primary analysis windows | outside support cells in primary analysis windows | note                                                                                                                                                                                     |
|----------|---------|--------------------------------------|--------------------|---------------|------------------|---------------------------|------------------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DeltaDTM | v1.1.1  | DeltaDTM v1.1.1 / DeltaDTM.jl v1.1.1 | EGM2008            | 30.000        | 10.000           | 254                       | 0                                              | 3320                                              | The 2024 paper documents v1.0; the downloaded v1.1 README documents the 30 m EGM2008 cap. Outside-support fill is kept distinct from class 255 and excluded from all land denominator s. |

Supplementary Table S3. Primary association, spatial interval and top-set statistics.

| Terrain metric               | n  | Spearman rho | Kendall tau | Two-sided label-permutation p | Top-set n | Observed overlap | Expected independent overlap | Sector bootstrap 2.5% | Sector bootstrap 97.5% | Sector-adjusted rank association | Within-sector permutation p |
|------------------------------|----|--------------|-------------|-------------------------------|-----------|------------------|------------------------------|-----------------------|------------------------|----------------------------------|-----------------------------|
| connected_reference_land_pct | 68 | -0.333       | -0.211      | 0.006                         | 14        | 1                | 2.882                        | -0.647                | 0.179                  | -0.492                           | 4.000e-04                   |
| below_reference_land_pct     | 68 | -0.309       | -0.188      | 0.009                         | 14        | 1                | 2.882                        | -0.624                | 0.191                  | -0.454                           | 8.999e-04                   |
| conditional_connectivity_pct | 68 | 0.108        | 0.072       | 0.379                         | 14        | 5                | 2.882                        | -0.276                | 0.322                  | 0.198                            | 0.420                       |
| connected_area_km2           | 68 | -0.341       | -0.208      | 0.005                         | 14        | 1                | 2.882                        | -0.626                | 0.208                  | -0.489                           | 2.000e-04                   |

Supplementary Table S4. COAST-RP return-period sensitivity with spatially constrained inference.

| Return period (years) | Terrain metric           | n  | Spearman rho | Kendall tau | Sector-adjusted rank association | Within-sector p | Top-20% overlap | Spatial-null lower-tail p |
|-----------------------|--------------------------|----|--------------|-------------|----------------------------------|-----------------|-----------------|---------------------------|
| 2                     | lowland_prevalence       | 68 | -0.335       | -0.205      | -0.446                           | 2.000e-04       | 0               | 0.050                     |
| 2                     | conditional_connectivity | 68 | 0.104        | 0.066       | 0.196                            | 0.410           | 5               | 0.984                     |

| Return period (years) | Terrain metric           | n  | Spearman rho | Kendall tau | Sector-adjusted rank association | Within-sector p | Top-20% overlap | Spatial-null lower-tail p |
|-----------------------|--------------------------|----|--------------|-------------|----------------------------------|-----------------|-----------------|---------------------------|
| 2                     | connected_lowland_share  | 68 | -0.359       | -0.226      | -0.483                           | 9.999e-05       | 0               | 0.086                     |
| 2                     | connected_area           | 68 | -0.369       | -0.228      | -0.481                           | 9.999e-05       | 0               | 0.086                     |
| 2                     | connected_window_share   | 68 | -0.369       | -0.228      | -0.481                           | 9.999e-05       | 0               | 0.081                     |
| 5                     | lowland_prevalence       | 68 | -0.318       | -0.193      | -0.451                           | 4.000e-04       | 1               | 0.176                     |
| 5                     | conditional_connectivity | 68 | 0.105        | 0.067       | 0.197                            | 0.408           | 5               | 0.988                     |
| 5                     | connected_lowland_share  | 68 | -0.341       | -0.216      | -0.488                           | 4.000e-04       | 1               | 0.263                     |
| 5                     | connected_area           | 68 | -0.349       | -0.215      | -0.485                           | 2.000e-04       | 1               | 0.255                     |
| 5                     | connected_window_share   | 68 | -0.349       | -0.215      | -0.485                           | 9.999e-05       | 1               | 0.259                     |
| 10                    | lowland_prevalence       | 68 | -0.309       | -0.188      | -0.454                           | 8.999e-04       | 1               | 0.173                     |
| 10                    | conditional_connectivity | 68 | 0.105        | 0.069       | 0.198                            | 0.420           | 5               | 0.987                     |
| 10                    | connected_lowland_share  | 68 | -0.333       | -0.211      | -0.492                           | 4.000e-04       | 1               | 0.257                     |
| 10                    | connected_area           | 68 | -0.341       | -0.208      | -0.489                           | 2.000e-04       | 1               | 0.255                     |
| 10                    | connected_window_share   | 68 | -0.341       | -0.208      | -0.489                           | 2.000e-04       | 1               | 0.258                     |
| 25                    | lowland_prevalence       | 68 | -0.304       | -0.183      | -0.454                           | 0.002           | 1               | 0.163                     |
| 25                    | conditional_connectivity | 68 | 0.106        | 0.074       | 0.198                            | 0.405           | 5               | 0.988                     |
| 25                    | connected_lowland_share  | 68 | -0.327       | -0.206      | -0.492                           | 5.000e-04       | 1               | 0.255                     |
| 25                    | connected_area           | 68 | -0.333       | -0.204      | -0.487                           | 9.999e-05       | 1               | 0.259                     |
| 25                    | connected_window_share   | 68 | -0.333       | -0.204      | -0.487                           | 9.999e-05       | 1               | 0.255                     |
| 50                    | lowland_prevalence       | 68 | -0.299       | -0.180      | -0.457                           | 0.002           | 1               | 0.172                     |
| 50                    | conditional_connectivity | 68 | 0.100        | 0.068       | 0.194                            | 0.446           | 5               | 0.987                     |
| 50                    | connected_lowland_share  | 68 | -0.323       | -0.203      | -0.495                           | 7.999e-04       | 1               | 0.254                     |
| 50                    | connected_area           | 68 | -0.328       | -0.200      | -0.489                           | 3.000e-04       | 1               | 0.252                     |
| 50                    | connected_window_share   | 68 | -0.328       | -0.200      | -0.489                           | 2.000e-04       | 1               | 0.251                     |

| Return period (years) | Terrain metric           | n  | Spearman rho | Kendall tau | Sector-adjusted rank association | Within-sector p | Top-20% overlap | Spatial-null lower-tail p |
|-----------------------|--------------------------|----|--------------|-------------|----------------------------------|-----------------|-----------------|---------------------------|
| 100                   | lowland_prevalence       | 68 | -0.290       | -0.177      | -0.448                           | 0.002           | 1               | 0.170                     |
| 100                   | conditional_connectivity | 68 | 0.095        | 0.066       | 0.191                            | 0.470           | 5               | 0.987                     |
| 100                   | connected_lowland_share  | 68 | -0.317       | -0.198      | -0.491                           | 7.999e-04       | 1               | 0.247                     |
| 100                   | connected_areas          | 68 | -0.322       | -0.196      | -0.484                           | 3.000e-04       | 1               | 0.248                     |
| 100                   | connected_window_share   | 68 | -0.322       | -0.196      | -0.484                           | 9.999e-05       | 1               | 0.250                     |

**Supplementary Table S5. Full 68-station primary source table.**

| Station ID           | Station              | Coastal sector                     | Latitude | Longitude | COAST-RP RP10 (m) | Match distance (km) | Below-2 m land share (%) | Conditional connection (%) | Connected land share (%) | Connected area (km2) |
|----------------------|----------------------|------------------------------------|----------|-----------|-------------------|---------------------|--------------------------|----------------------------|--------------------------|----------------------|
| avonmouth-po60-uk    | Avonmouth Po60 Uk    | western-English-Channel            | 51.511   | -2.715    | 8.787             | 0.138               | 0.455                    | 97.123                     | 0.442                    | 0.240                |
| portbury-p341-uk     | Portbury P341 Uk     | western-English-Channel            | 51.500   | -2.728    | 8.738             | 0.412               | 0.383                    | 97.611                     | 0.374                    | 0.218                |
| newport-po57-uk      | Newport Po57 Uk      | western-English-Channel            | 51.550   | -2.987    | 8.602             | 0.711               | 0.164                    | 90.528                     | 0.148                    | 0.102                |
| hinkleypoint-po03-uk | Hinkleypoint Po03 Uk | western-English-Channel            | 51.215   | -3.134    | 7.734             | 0.608               | 0.109                    | 100.000                    | 0.109                    | 0.043                |
| saint-malo           | Saint Malo France    | western-English-Channel            | 48.641   | -2.028    | 7.428             | 0.129               | 0.487                    | 68.459                     | 0.333                    | 0.183                |
| heysham-po50-uk      | Heysham Po50 Uk      | British-North-Sea                  | 54.032   | -2.920    | 6.624             | 0.153               | 1.124                    | 99.103                     | 1.114                    | 0.371                |
| sthelier-po74-uk     | Sthelier Po74 Uk     | western-English-Channel            | 49.183   | -2.117    | 6.501             | 0.113               | 0.228                    | 100.000                    | 0.228                    | 0.125                |
| dieppe               | Dieppe France        | eastern-Channel-southern-North-Sea | 49.929   | 1.085     | 6.143             | 0.537               | 0.288                    | 85.823                     | 0.248                    | 0.134                |
| mumbles-p932-uk      | Mumbles P932 Uk      | western-English-Channel            | 51.570   | -3.975    | 5.825             | 0.111               | 0.207                    | 93.830                     | 0.195                    | 0.045                |
| ilfracombe-po61-uk   | Ilfracombe Po61 Uk   | western-English-Channel            | 51.211   | -4.111    | 5.765             | 0.789               | 0.171                    | 100.000                    | 0.171                    | 0.082                |
| boulogne-sur-mer     | Boulogne Sur Mer     | eastern-Channel-southern-North-Sea | 50.727   | 1.578     | 5.713             | 0.801               | 0.944                    | 99.795                     | 0.942                    | 0.441                |

| Station ID           | Station              | Coastal sector                     | Latitude | Longitude | COAST-RP RP10 (m) | Match distance (km) | Below-2 m land share (%) | Conditional connection (%) | Connected land share (%) | Connected area (km2) |
|----------------------|----------------------|------------------------------------|----------|-----------|-------------------|---------------------|--------------------------|----------------------------|--------------------------|----------------------|
| gladstone-p234-uk    | Gladstone P234 Uk    | British-North-Sea                  | 53.450   | -3.018    | 5.577             | 0.724               | 0.309                    | 66.396                     | 0.205                    | 0.139                |
| workington-p217-uk   | Workington P217 Uk   | British-North-Sea                  | 54.651   | -3.568    | 5.426             | 1.113               | 0.819                    | 100.000                    | 0.819                    | 0.362                |
| sheerness-p015-uk    | Sheerness            | eastern-Channel-southern-North-Sea | 51.442   | 0.743     | 5.256             | 0.493               | 45.712                   | 93.412                     | 42.700                   | 20.364               |
| llandudno-p007-uk    | Llandudno P007 Uk    | British-North-Sea                  | 53.332   | -3.825    | 5.041             | 0.588               | 0.826                    | 42.320                     | 0.349                    | 0.106                |
| roscoff              | Roscoff France       | western-English-Channel            | 48.718   | -3.966    | 4.857             | 0.619               | 0.911                    | 95.320                     | 0.869                    | 0.244                |
| calais               | Calais France        | eastern-Channel-southern-North-Sea | 50.969   | 1.868     | 4.700             | 1.668               | 37.398                   | 76.721                     | 28.692                   | 14.955               |
| cuxhaven-germany     | Cuxhaven Germany     | Denmark-Norway                     | 53.867   | 8.717     | 4.696             | 0.531               | 61.332                   | 97.032                     | 59.512                   | 28.550               |
| le-havre             | Le Havre France      | eastern-Channel-southern-North-Sea | 49.482   | 0.106     | 4.643             | 1.113               | 3.597                    | 92.096                     | 3.312                    | 1.401                |
| dover-po12-uk        | Dover PO12 Uk        | eastern-Channel-southern-North-Sea | 51.117   | 1.318     | 4.618             | 0.508               | 0.285                    | 93.878                     | 0.268                    | 0.124                |
| dunkerque            | Dunkerque France     | eastern-Channel-southern-North-Sea | 51.048   | 2.367     | 4.548             | 1.002               | 34.580                   | 88.407                     | 30.571                   | 16.083               |
| immingham-po26-uk    | Immingham Po26 Uk    | British-North-Sea                  | 53.631   | -0.186    | 4.528             | 0.487               | 20.142                   | 2.014                      | 0.406                    | 0.249                |
| milfordhaven-po56-uk | Milfordhaven Po56 Uk | Atlantic-Celtic                    | 51.702   | -5.014    | 4.376             | 0.186               | 0.573                    | 82.337                     | 0.472                    | 0.367                |
| brest                | Brest                | western-English-Channel            | 48.383   | -4.500    | 4.313             | 0.833               | 0.459                    | 61.245                     | 0.281                    | 0.176                |
| newhaven-po11-uk     | Newhaven PO11 Uk     | eastern-Channel-southern-North-Sea | 50.782   | 0.057     | 4.261             | 0.540               | 5.849                    | 64.710                     | 3.785                    | 2.031                |
| delfzijl-del-nl      | Delfzijl Del NL      | Dutch-German-Bight                 | 53.326   | 6.933     | 4.094             | 0.294               | 97.003                   | 99.692                     | 96.704                   | 61.303               |
| le-conquet           | Le Conquet France    | western-English-Channel            | 48.359   | -4.781    | 4.027             | 0.374               | 0.338                    | 90.053                     | 0.304                    | 0.109                |

| Station ID             | Station                | Coastal sector                     | Latitude | Longitude | COAST-RP RP10 (m) | Match distance (km) | Below-2 m land share (%) | Conditional connection (%) | Connected land share (%) | Connected area (km2) |
|------------------------|------------------------|------------------------------------|----------|-----------|-------------------|---------------------|--------------------------|----------------------------|--------------------------|----------------------|
| harwich-p022-uk        | Harwich Po22 Uk        | eastern-Channel-southern-North-Sea | 51.947   | 1.284     | 3.861             | 0.477               | 24.490                   | 90.941                     | 22.272                   | 12.966               |
| felixstowe-p204-uk     | Felixstowe P204 Uk     | eastern-Channel-southern-North-Sea | 51.957   | 1.348     | 3.750             | 0.806               | 27.837                   | 96.247                     | 26.793                   | 10.628               |
| cromer-p183-uk         | Cromer P183 Uk         | British-North-Sea                  | 52.934   | 1.302     | 3.740             | 0.436               | 0.315                    | 100.000                    | 0.315                    | 0.142                |
| cherbourg              | Cherbourg France       | western-English-Channel            | 49.651   | -1.635    | 3.674             | 0.180               | 1.470                    | 91.840                     | 1.351                    | 0.779                |
| porterin-p919-uk       | Porterin P919 Uk       | British-North-Sea                  | 54.085   | -4.768    | 3.640             | 0.445               | 2.170                    | 100.000                    | 2.170                    | 0.723                |
| leith-p034-uk          | Leith Po34 Uk          | British-North-Sea                  | 55.990   | -3.182    | 3.589             | 0.519               | 0.293                    | 82.824                     | 0.243                    | 0.088                |
| barmouth-p923-uk       | Barmouth P923 Uk       | British-North-Sea                  | 52.719   | -4.045    | 3.525             | 1.544               | 6.693                    | 87.946                     | 5.886                    | 3.535                |
| whitby-p174-uk         | Whitby P174 Uk         | British-North-Sea                  | 54.490   | -0.614    | 3.431             | 0.684               | 0.449                    | 61.040                     | 0.274                    | 0.135                |
| esbjerg-130121-denmark | Esbjerg 130121 Denmark | Denmark-Norway                     | 55.467   | 8.433     | 3.399             | 1.096               | 9.225                    | 88.797                     | 8.192                    | 4.926                |
| hoekvanholla-hvh-nl    | Hoek van Holland       | Dutch-German-Bight                 | 51.977   | 4.120     | 3.373             | 0.432               | 48.135                   | 5.484                      | 2.640                    | 1.411                |
| stmarys-p231-uk        | Stmarys P231 Uk        | Atlantic-Celtic                    | 49.918   | -6.316    | 3.371             | 0.469               | 18.971                   | 96.030                     | 18.218                   | 2.911                |
| newlyn-p001-uk         | Newlyn                 | Atlantic-Celtic                    | 50.102   | -5.542    | 3.346             | 0.899               | 0.454                    | 98.367                     | 0.446                    | 0.276                |
| holyhead-p054-uk       | Holyhead Po54 Uk       | British-North-Sea                  | 53.308   | -4.631    | 3.315             | 0.641               | 7.956                    | 71.688                     | 5.703                    | 2.190                |
| northshields-p032-uk   | Northshields Po32 Uk   | British-North-Sea                  | 55.007   | -1.440    | 3.302             | 0.936               | 0.199                    | 76.421                     | 0.152                    | 0.103                |
| saint-nazaire          | Saint Nazaire France   | western-English-Channel            | 47.267   | -2.202    | 3.275             | 0.513               | 10.684                   | 5.465                      | 0.584                    | 0.331                |
| ullapool-p043-uk       | Ullapool Po43 Uk       | Atlantic-Celtic                    | 57.895   | -5.158    | 3.256             | 0.491               | 0.279                    | 98.970                     | 0.276                    | 0.219                |
| saint-gildas           | Saint Gildas France    | western-English-Channel            | 47.140   | -2.246    | 3.243             | 0.696               | 4.497                    | 99.308                     | 4.466                    | 0.652                |
| kinlochbervie-p918-uk  | Kinlochbervie P918 Uk  | Atlantic-Celtic                    | 58.457   | -5.050    | 3.151             | 2.314               | 0.730                    | 94.512                     | 0.690                    | 0.490                |
| le-crouesty            | Le Crouesty France     | western-English-Channel            | 47.543   | -2.895    | 3.131             | 0.199               | 4.551                    | 35.605                     | 1.620                    | 0.439                |

| Station ID              | Station             | Coastal sector          | Latitude | Longitude | COAST-RP RP10 (m) | Match distance (km) | Below-2 m land share (%) | Conditional connection (%) | Connected land share (%) | Connected area (km2) |
|-------------------------|---------------------|-------------------------|----------|-----------|-------------------|---------------------|--------------------------|----------------------------|--------------------------|----------------------|
| tobermory-p223-uk       | Tobermory P223 Uk   | Atlantic-Celtic         | 56.623   | -6.064    | 3.091             | 0.454               | 0.530                    | 99.448                     | 0.527                    | 0.284                |
| devonport-p002-uk       | Devonport P002 Uk   | western-English-Channel | 50.368   | -4.185    | 3.048             | 0.253               | 0.980                    | 95.605                     | 0.937                    | 0.615                |
| concarneau              | Concarneau France   | western-English-Channel | 47.874   | -3.907    | 3.035             | 1.373               | 0.955                    | 98.821                     | 0.944                    | 0.644                |
| port-tudy               | Port Tudy France    | western-English-Channel | 47.644   | -3.446    | 3.000             | 0.307               | 1.247                    | 99.666                     | 1.243                    | 0.192                |
| fishguard-p055-uk       | Fishguard P055 Uk   | British-North-Sea       | 52.014   | -4.983    | 2.998             | 0.642               | 0.227                    | 78.340                     | 0.178                    | 0.072                |
| morayfirth-p207-uk      | Morayfirth P207 Uk  | British-North-Sea       | 57.599   | -4.002    | 2.990             | 0.704               | 4.139                    | 92.637                     | 3.834                    | 1.448                |
| stornoway-p042-uk       | Stornoway P042 Uk   | Atlantic-Celtic         | 58.208   | -6.388    | 2.980             | 1.965               | 2.262                    | 84.451                     | 1.911                    | 1.343                |
| portpatrick-p063-uk     | Portpatrick P063 Uk | Atlantic-Celtic         | 54.843   | -5.120    | 2.887             | 0.779               | 0.548                    | 99.669                     | 0.546                    | 0.248                |
| aberdeen-p038-uk        | Aberdeen            | British-North-Sea       | 57.143   | -2.075    | 2.822             | 1.264               | 1.170                    | 83.981                     | 0.982                    | 0.523                |
| portsmouth-p008-uk      | Portsmouth P008 Uk  | western-English-Channel | 50.803   | -1.112    | 2.767             | 0.689               | 16.896                   | 88.826                     | 15.008                   | 9.510                |
| millport-p049-uk        | Millport P049 Uk    | British-North-Sea       | 55.750   | -4.906    | 2.717             | 0.664               | 1.816                    | 84.633                     | 1.537                    | 0.688                |
| denhelder-hel-nl        | Denhelder Hel NL    | Dutch-German-Bight      | 52.965   | 4.746     | 2.636             | 0.627               | 72.950                   | 98.327                     | 71.730                   | 20.855               |
| bangor-p662-uk          | Bangor P662 Uk      | Atlantic-Celtic         | 54.665   | -5.670    | 2.615             | 2.811               | 1.592                    | 100.000                    | 1.592                    | 0.913                |
| wick-p035-uk            | Wick P035 Uk        | British-North-Sea       | 58.441   | -3.086    | 2.560             | 2.044               | 1.580                    | 98.213                     | 1.552                    | 0.864                |
| malin-head-834a-ireland | Malin Head 834a     | Atlantic-Celtic         | 55.367   | -7.333    | 2.555             | 0.344               | 1.191                    | 99.421                     | 1.184                    | 0.419                |
| lowestoft-p024-uk       | Lowestoft P024 Uk   | British-North-Sea       | 52.473   | 1.751     | 2.379             | 0.559               | 12.097                   | 11.707                     | 1.416                    | 0.670                |
| portrush-p935-uk        | Portrush P935 Uk    | Atlantic-Celtic         | 55.207   | -6.657    | 2.005             | 0.451               | 1.115                    | 82.627                     | 0.922                    | 0.389                |
| weymouth-p991-uk        | Weymouth P991 Uk    | western-English-Channel | 50.608   | -2.448    | 1.902             | 0.839               | 5.139                    | 59.202                     | 3.043                    | 1.350                |
| lerwick-p041-uk         | Lerwick P041 Uk     | British-North-Sea       | 60.150   | -1.133    | 1.508             | 0.094               | 2.493                    | 93.779                     | 2.338                    | 1.331                |
| portellen-p202-uk       | Portellen P202 Uk   | Atlantic-Celtic         | 55.627   | -6.190    | 1.437             | 0.069               | 0.963                    | 99.287                     | 0.956                    | 0.562                |

| Station ID          | Station             | Coastal sector          | Latitude | Longitude | COAST-RP RP10 (m) | Match distance (km) | Below-2 m land share (%) | Conditional connection (%) | Connected land share (%) | Connected area (km2) |
|---------------------|---------------------|-------------------------|----------|-----------|-------------------|---------------------|--------------------------|----------------------------|--------------------------|----------------------|
| bournemouth-p988-uk | Bournemouth P988 Uk | western-English-Channel | 50.714   | -1.875    | 1.395             | 0.825               | 0.883                    | 81.183                     | 0.717                    | 0.375                |
| tregde-006-norway   | Tregde 006 Norway   | Dutch-German-Bight      | 58.000   | 7.567     | 0.999             | 0.035               | 6.605                    | 86.397                     | 5.707                    | 2.743                |

**Supplementary Table S6. Window support and alternative connected-window denominator.**

| Station                | Nominal window area (km2) | Represented land area (km2) | Represented land (%) | Below-2 m area (km2) | Connected area (km2) | Connected window share (%) |
|------------------------|---------------------------|-----------------------------|----------------------|----------------------|----------------------|----------------------------|
| Aberdeen               | 100.569                   | 53.285                      | 52.983               | 0.623                | 0.523                | 0.520                      |
| Avonmouth P060 Uk      | 100.461                   | 54.457                      | 54.207               | 0.248                | 0.240                | 0.239                      |
| Bangor P662 Uk         | 100.642                   | 57.315                      | 56.950               | 0.913                | 0.913                | 0.907                      |
| Barmouth P923 Uk       | 100.601                   | 60.045                      | 59.687               | 4.019                | 3.535                | 3.513                      |
| Boulogne Sur Mer       | 100.703                   | 46.826                      | 46.499               | 0.442                | 0.441                | 0.438                      |
| Bournemouth P988 Uk    | 100.436                   | 52.338                      | 52.111               | 0.462                | 0.375                | 0.374                      |
| Brest                  | 100.595                   | 62.636                      | 62.266               | 0.288                | 0.176                | 0.175                      |
| Calais France          | 100.672                   | 52.124                      | 51.776               | 19.493               | 14.955               | 14.856                     |
| Cherbourg France       | 100.481                   | 57.661                      | 57.384               | 0.848                | 0.779                | 0.775                      |
| Concarneau France      | 100.554                   | 68.207                      | 67.831               | 0.652                | 0.644                | 0.640                      |
| Cromer P183 Uk         | 100.666                   | 45.204                      | 44.905               | 0.142                | 0.142                | 0.141                      |
| Cuxhaven Germany       | 100.683                   | 47.974                      | 47.648               | 29.423               | 28.550               | 28.356                     |
| Delfzijl Del Nl        | 100.495                   | 63.392                      | 63.080               | 61.492               | 61.303               | 61.001                     |
| Denhelder Hel Nl       | 100.594                   | 29.074                      | 28.902               | 21.209               | 20.855               | 20.731                     |
| Devonport P002 Uk      | 100.581                   | 65.644                      | 65.265               | 0.644                | 0.615                | 0.612                      |
| Dieppe France          | 100.507                   | 54.076                      | 53.803               | 0.156                | 0.134                | 0.133                      |
| Dover P012 Uk          | 100.448                   | 46.321                      | 46.114               | 0.132                | 0.124                | 0.123                      |
| Dunkerque France       | 100.599                   | 52.607                      | 52.294               | 18.192               | 16.083               | 15.987                     |
| Esbjerg 130121 Denmark | 100.496                   | 60.130                      | 59.833               | 5.547                | 4.926                | 4.901                      |
| Felixstowe P204 Uk     | 100.624                   | 39.669                      | 39.423               | 11.043               | 10.628               | 10.563                     |
| Fishguard P055 Uk      | 100.496                   | 40.629                      | 40.428               | 0.092                | 0.072                | 0.072                      |
| Gladstone P234 Uk      | 100.573                   | 67.842                      | 67.455               | 0.210                | 0.139                | 0.138                      |
| Harwich P022 Uk        | 100.646                   | 58.218                      | 57.845               | 14.258               | 12.966               | 12.883                     |
| Heysham P050 Uk        | 100.652                   | 33.354                      | 33.138               | 0.375                | 0.371                | 0.369                      |
| Hinkleypoint P003 Uk   | 100.527                   | 39.514                      | 39.307               | 0.043                | 0.043                | 0.043                      |
| Hoek van Holland       | 100.579                   | 53.453                      | 53.145               | 25.730               | 1.411                | 1.403                      |
| Holyhead P054 Uk       | 100.631                   | 38.395                      | 38.155               | 3.055                | 2.190                | 2.176                      |
| Ilfracombe P061 Uk     | 100.536                   | 48.265                      | 48.007               | 0.082                | 0.082                | 0.082                      |

| Station              | Nominal window area (km2) | Represented land area (km2) | Represented land (%) | Below-2 m area (km2) | Connected area (km2) | Connected window share (%) |
|----------------------|---------------------------|-----------------------------|----------------------|----------------------|----------------------|----------------------------|
| Immingham P026 Uk    | 100.698                   | 61.511                      | 61.085               | 12.389               | 0.249                | 0.248                      |
| Kinlochbervi P918 Uk | 100.552                   | 71.053                      | 70.664               | 0.519                | 0.490                | 0.488                      |
| Le Conquet France    | 100.643                   | 35.849                      | 35.620               | 0.121                | 0.109                | 0.108                      |
| Le Crouesty France   | 100.565                   | 27.100                      | 26.947               | 1.233                | 0.439                | 0.437                      |
| Le Havre France      | 100.627                   | 42.298                      | 42.034               | 1.521                | 1.401                | 1.392                      |
| Leith P034 Uk        | 100.551                   | 36.078                      | 35.880               | 0.106                | 0.088                | 0.087                      |
| Lerwick P041 Uk      | 100.621                   | 56.926                      | 56.574               | 1.419                | 1.331                | 1.323                      |
| Llandudno P007 Uk    | 100.574                   | 30.347                      | 30.174               | 0.251                | 0.106                | 0.105                      |
| Lowestoft P024 Uk    | 100.599                   | 47.276                      | 46.995               | 5.719                | 0.670                | 0.666                      |
| Malin Head 834a      | 100.486                   | 35.379                      | 35.208               | 0.421                | 0.419                | 0.417                      |
| Milfordhaven P056 Uk | 100.617                   | 77.703                      | 77.226               | 0.445                | 0.367                | 0.364                      |
| Millport P049 Uk     | 100.561                   | 44.739                      | 44.490               | 0.813                | 0.688                | 0.684                      |
| Morayfirth P207 Uk   | 100.660                   | 37.755                      | 37.507               | 1.563                | 1.448                | 1.438                      |
| Mumbles P932 Uk      | 100.524                   | 23.148                      | 23.027               | 0.048                | 0.045                | 0.045                      |
| Newhaven P011 Uk     | 100.487                   | 53.674                      | 53.414               | 3.139                | 2.031                | 2.022                      |
| Newlyn               | 100.544                   | 61.934                      | 61.599               | 0.281                | 0.276                | 0.275                      |
| Newport P057 Uk      | 100.568                   | 68.877                      | 68.488               | 0.113                | 0.102                | 0.101                      |
| Northshields P032 Uk | 100.596                   | 67.751                      | 67.350               | 0.135                | 0.103                | 0.102                      |
| Port Tudy France     | 100.581                   | 15.418                      | 15.329               | 0.192                | 0.192                | 0.190                      |
| Portbury P341 Uk     | 100.485                   | 58.347                      | 58.066               | 0.224                | 0.218                | 0.217                      |
| Portellen P202 Uk    | 100.615                   | 58.849                      | 58.490               | 0.566                | 0.562                | 0.559                      |
| Porterin P919 Uk     | 100.705                   | 33.311                      | 33.078               | 0.723                | 0.723                | 0.718                      |
| Portpatrick P063 Uk  | 100.469                   | 45.421                      | 45.209               | 0.249                | 0.248                | 0.247                      |
| Portrush P935 Uk     | 100.625                   | 42.190                      | 41.928               | 0.471                | 0.389                | 0.386                      |
| Portsmouth P008 Uk   | 100.540                   | 63.365                      | 63.024               | 10.706               | 9.510                | 9.459                      |
| Roscoff France       | 100.547                   | 28.040                      | 27.887               | 0.256                | 0.244                | 0.242                      |
| Saint Gildas France  | 100.489                   | 14.602                      | 14.531               | 0.657                | 0.652                | 0.649                      |
| Saint Malo France    | 100.496                   | 55.045                      | 54.774               | 0.268                | 0.183                | 0.182                      |
| Saint Nazaire France | 100.671                   | 56.653                      | 56.275               | 6.053                | 0.331                | 0.329                      |
| Sheerness            | 100.613                   | 47.691                      | 47.401               | 21.801               | 20.364               | 20.240                     |
| Sthelier P074 Uk     | 100.631                   | 55.017                      | 54.672               | 0.125                | 0.125                | 0.125                      |
| Stmarys P231 Uk      | 100.530                   | 15.979                      | 15.895               | 3.031                | 2.911                | 2.896                      |
| Stornoway P042 Uk    | 100.525                   | 70.287                      | 69.920               | 1.590                | 1.343                | 1.336                      |
| Tobermory P223 Uk    | 100.610                   | 53.852                      | 53.526               | 0.285                | 0.284                | 0.282                      |
| Tregde 006 Norway    | 100.537                   | 48.073                      | 47.816               | 3.175                | 2.743                | 2.729                      |
| Ullapool P043 Uk     | 100.665                   | 79.382                      | 78.857               | 0.221                | 0.219                | 0.218                      |
| Weymouth P991 Uk     | 100.663                   | 44.385                      | 44.093               | 2.281                | 1.350                | 1.342                      |

| Station            | Nominal window area (km2) | Represented land area (km2) | Represented land (%) | Below-2 m area (km2) | Connected area (km2) | Connected window share (%) |
|--------------------|---------------------------|-----------------------------|----------------------|----------------------|----------------------|----------------------------|
| Whitby P174 Uk     | 100.533                   | 49.430                      | 49.168               | 0.222                | 0.135                | 0.135                      |
| Wick P035 Uk       | 100.597                   | 55.683                      | 55.353               | 0.880                | 0.864                | 0.859                      |
| Workington P217 Uk | 100.677                   | 44.181                      | 43.884               | 0.362                | 0.362                | 0.359                      |

**Supplementary Table S7. Denominator-adjusted associations.**

| terrain metric               | terrain metric label     | n sites | raw spearman water vs metric | partial spearman controlling represented land fraction | spearman metric vs represented land fraction | spearman water vs represented land fraction |
|------------------------------|--------------------------|---------|------------------------------|--------------------------------------------------------|----------------------------------------------|---------------------------------------------|
| below_reference_land_pct     | lowland_prevalence       | 68      | -0.309                       | -0.310                                                 | -0.098                                       | 0.003                                       |
| conditional_connectivity_pct | conditional_connectivity | 68      | 0.105                        | 0.106                                                  | -0.128                                       | 0.003                                       |
| connected_reference_land_pct | connected_lowland_share  | 68      | -0.333                       | -0.336                                                 | -0.137                                       | 0.003                                       |
| connected_area_km2           | connected_area           | 68      | -0.341                       | -0.342                                                 | 0.051                                        | 0.003                                       |
| connected_window_pct         | connected_window_share   | 68      | -0.341                       | -0.342                                                 | 0.051                                        | 0.003                                       |

**Supplementary Table S8. One-at-a-time full-sample robustness grid.**

| Setting                                  | Terrain metric               | n  | Spearman rho | Kendall tau | Two-sided p | Terrain-rank rho vs primary | Top-20% Jaccard vs primary | Top-20% membership changes |
|------------------------------------------|------------------------------|----|--------------|-------------|-------------|-----------------------------|----------------------------|----------------------------|
| primary:10km, 2m,4n,adaptive ,match<=6km | connected_reference_land_pct | 68 | -0.333       | -0.211      | 0.006       | 1.000                       | 1.000                      | 0                          |
| primary:10km, 2m,4n,adaptive ,match<=6km | below_reference_land_pct     | 68 | -0.309       | -0.188      | 0.009       | 1.000                       | 1.000                      | 0                          |
| primary:10km, 2m,4n,adaptive ,match<=6km | conditional_connectivity_pct | 68 | 0.108        | 0.072       | 0.379       | 1.000                       | 1.000                      | 0                          |
| primary:10km, 2m,4n,adaptive ,match<=6km | connected_area_km2           | 68 | -0.341       | -0.208      | 0.005       | 1.000                       | 1.000                      | 0                          |
| window:5km                               | connected_reference_land_pct | 68 | -0.410       | -0.273      | 6.999e-04   | 0.895                       | 0.556                      | 8                          |
| window:5km                               | below_reference_land_pct     | 68 | -0.333       | -0.217      | 0.006       | 0.932                       | 0.647                      | 6                          |
| window:5km                               | conditional_connectivity_pct | 68 | -0.012       | -0.030      | 0.926       | 0.740                       | 0.333                      | 14                         |
| window:5km                               | connected_area_km2           | 68 | -0.425       | -0.279      | 2.000e-04   | 0.898                       | 0.556                      | 8                          |
| window:20km                              | connected_reference_land_pct | 68 | -0.388       | -0.267      | 0.001       | 0.910                       | 0.750                      | 4                          |

| Setting                      | Terrain metric               | n  | Spearman rho | Kendall tau | Two-sided p | Terrain-rank rho vs primary | Top-20% Jaccard vs primary | Top-20% membership changes |
|------------------------------|------------------------------|----|--------------|-------------|-------------|-----------------------------|----------------------------|----------------------------|
| window:20km                  | below_reference_land_pct     | 68 | -0.317       | -0.212      | 0.010       | 0.916                       | 1.000                      | 0                          |
| window:20km                  | conditional_connectivity_pct | 68 | 0.032        | 0.018       | 0.802       | 0.663                       | 0.400                      | 12                         |
| window:20km                  | connected_area_km2           | 68 | -0.407       | -0.277      | 6.999e-04   | 0.890                       | 0.647                      | 6                          |
| threshold:1m                 | connected_reference_land_pct | 68 | -0.418       | -0.271      | 3.000e-04   | 0.844                       | 0.750                      | 4                          |
| threshold:1m                 | below_reference_land_pct     | 68 | -0.360       | -0.239      | 0.002       | 0.950                       | 0.867                      | 2                          |
| threshold:1m                 | conditional_connectivity_pct | 68 | 0.040        | 0.026       | 0.744       | 0.778                       | 0.647                      | 6                          |
| threshold:1m                 | connected_area_km2           | 68 | -0.422       | -0.273      | 4.000e-04   | 0.844                       | 0.750                      | 4                          |
| threshold:3m                 | connected_reference_land_pct | 68 | -0.295       | -0.177      | 0.014       | 0.938                       | 0.750                      | 4                          |
| threshold:3m                 | below_reference_land_pct     | 68 | -0.238       | -0.143      | 0.050       | 0.975                       | 0.867                      | 2                          |
| threshold:3m                 | conditional_connectivity_pct | 68 | 0.052        | 0.042       | 0.674       | 0.703                       | 0.556                      | 8                          |
| threshold:3m                 | connected_area_km2           | 68 | -0.292       | -0.179      | 0.016       | 0.936                       | 0.750                      | 4                          |
| neighbours:8                 | connected_reference_land_pct | 68 | -0.320       | -0.201      | 0.008       | 0.961                       | 0.647                      | 6                          |
| neighbours:8                 | below_reference_land_pct     | 68 | -0.309       | -0.188      | 0.011       | 1.000                       | 1.000                      | 0                          |
| neighbours:8                 | conditional_connectivity_pct | 68 | -0.001       | 0.003       | 0.992       | 0.900                       | 0.867                      | 2                          |
| neighbours:8                 | connected_area_km2           | 68 | -0.329       | -0.203      | 0.005       | 0.967                       | 0.647                      | 6                          |
| seed:local-ocean-only        | connected_reference_land_pct | 65 | -0.394       | -0.258      | 0.002       | 0.781                       | 0.733                      | 4                          |
| seed:local-ocean-only        | below_reference_land_pct     | 65 | -0.294       | -0.184      | 0.019       | 1.000                       | 1.000                      | 0                          |
| seed:local-ocean-only        | conditional_connectivity_pct | 65 | -0.058       | -0.045      | 0.647       | 0.862                       | 0.733                      | 4                          |
| seed:local-ocean-only        | connected_area_km2           | 65 | -0.423       | -0.272      | 4.000e-04   | 0.780                       | 0.733                      | 4                          |
| uniform-elevation-offset:-1m | connected_reference_land_pct | 68 | -0.295       | -0.177      | 0.015       | 0.938                       | 0.750                      | 4                          |
| uniform-elevation-offset:-1m | below_reference_land_pct     | 68 | -0.238       | -0.143      | 0.053       | 0.975                       | 0.867                      | 2                          |

| Setting                        | Terrain metric               | n  | Spearman rho | Kendall tau | Two-sided p | Terrain-rank rho vs primary | Top-20% Jaccard vs primary | Top-20% membership changes |
|--------------------------------|------------------------------|----|--------------|-------------|-------------|-----------------------------|----------------------------|----------------------------|
| uniform-elevation-offset:-1m   | conditional_connectivity_pct | 68 | 0.052        | 0.042       | 0.659       | 0.703                       | 0.556                      | 8                          |
| uniform-elevation-offset:-1m   | connected_area_km2           | 68 | -0.292       | -0.179      | 0.015       | 0.936                       | 0.750                      | 4                          |
| uniform-elevation-offset:-0.5m | connected_reference_land_pct | 68 | -0.310       | -0.191      | 0.010       | 0.949                       | 0.750                      | 4                          |
| uniform-elevation-offset:-0.5m | below_reference_land_pct     | 68 | -0.277       | -0.169      | 0.022       | 0.990                       | 1.000                      | 0                          |
| uniform-elevation-offset:-0.5m | conditional_connectivity_pct | 68 | 0.066        | 0.047       | 0.592       | 0.802                       | 0.867                      | 2                          |
| uniform-elevation-offset:-0.5m | connected_area_km2           | 68 | -0.308       | -0.188      | 0.010       | 0.956                       | 0.750                      | 4                          |
| uniform-elevation-offset:+0.5m | connected_reference_land_pct | 68 | -0.394       | -0.260      | 7.999e-04   | 0.938                       | 0.750                      | 4                          |
| uniform-elevation-offset:+0.5m | below_reference_land_pct     | 68 | -0.320       | -0.200      | 0.008       | 0.993                       | 1.000                      | 0                          |
| uniform-elevation-offset:+0.5m | conditional_connectivity_pct | 68 | -0.015       | -0.014      | 0.907       | 0.837                       | 0.556                      | 8                          |
| uniform-elevation-offset:+0.5m | connected_area_km2           | 68 | -0.396       | -0.250      | 0.001       | 0.940                       | 0.867                      | 2                          |
| uniform-elevation-offset:+1m   | connected_reference_land_pct | 68 | -0.418       | -0.271      | 2.000e-04   | 0.844                       | 0.750                      | 4                          |
| uniform-elevation-offset:+1m   | below_reference_land_pct     | 68 | -0.360       | -0.239      | 0.002       | 0.950                       | 0.867                      | 2                          |
| uniform-elevation-offset:+1m   | conditional_connectivity_pct | 68 | 0.040        | 0.026       | 0.743       | 0.778                       | 0.647                      | 6                          |
| uniform-elevation-offset:+1m   | connected_area_km2           | 68 | -0.422       | -0.273      | 4.000e-04   | 0.844                       | 0.750                      | 4                          |
| COAST-match: <=1km             | connected_reference_land_pct | 56 | -0.420       | -0.273      | 0.001       | 1.000                       | 1.000                      | 0                          |
| COAST-match: <=1km             | below_reference_land_pct     | 56 | -0.376       | -0.232      | 0.004       | 1.000                       | 1.000                      | 0                          |
| COAST-match: <=1km             | conditional_connectivity_pct | 56 | 0.148        | 0.098       | 0.279       | 1.000                       | 1.000                      | 0                          |

| Setting                           | Terrain metric               | n  | Spearman rho | Kendall tau | Two-sided p | Terrain-rank rho vs primary | Top-20% Jaccard vs primary | Top-20% membership changes |
|-----------------------------------|------------------------------|----|--------------|-------------|-------------|-----------------------------|----------------------------|----------------------------|
| COAST-match: <=1km                | connected_area_km2           | 56 | -0.414       | -0.264      | 0.002       | 1.000                       | 1.000                      | 0                          |
| COAST-match: <=2km                | connected_reference_land_pct | 65 | -0.334       | -0.210      | 0.005       | 1.000                       | 1.000                      | 0                          |
| COAST-match: <=2km                | below_reference_land_pct     | 65 | -0.315       | -0.191      | 0.012       | 1.000                       | 1.000                      | 0                          |
| COAST-match: <=2km                | conditional_connectivity_pct | 65 | 0.164        | 0.109       | 0.199       | 1.000                       | 1.000                      | 0                          |
| COAST-match: <=2km                | connected_area_km2           | 65 | -0.335       | -0.202      | 0.007       | 1.000                       | 1.000                      | 0                          |
| COAST-match:distance-unrestricted | connected_reference_land_pct | 74 | -0.311       | -0.196      | 0.007       | 1.000                       | 1.000                      | 0                          |
| COAST-match:distance-unrestricted | below_reference_land_pct     | 74 | -0.279       | -0.163      | 0.015       | 1.000                       | 1.000                      | 0                          |
| COAST-match:distance-unrestricted | conditional_connectivity_pct | 74 | 0.116        | 0.081       | 0.324       | 1.000                       | 1.000                      | 0                          |
| COAST-match:distance-unrestricted | connected_area_km2           | 74 | -0.324       | -0.198      | 0.006       | 1.000                       | 1.000                      | 0                          |

**Supplementary Table S9. Spatially constrained top-k overlap null curves, with nonspatial hypergeometric comparators.**

| top share pct | n sites | top n | observed overlap | sector stratified null median | sector stratified null q025 | sector stratified null q975 | sector stratified lower tail p | nonspatial hypergeometric mean | nonspatial hypergeometric q025 | nonspatial hypergeometric q975 |
|---------------|---------|-------|------------------|-------------------------------|-----------------------------|-----------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| 10            | 68      | 7     | 0                | 0.000                         | 0.000                       | 0.000                       | 1.000                          | 0.721                          | 0.000                          | 2.000                          |
| 15            | 68      | 11    | 0                | 1.000                         | 0.000                       | 3.000                       | 0.140                          | 1.779                          | 0.000                          | 4.000                          |
| 20            | 68      | 14    | 1                | 2.000                         | 0.000                       | 4.000                       | 0.257                          | 2.882                          | 0.000                          | 6.000                          |
| 25            | 68      | 17    | 2                | 4.000                         | 2.000                       | 6.000                       | 0.133                          | 4.250                          | 1.000                          | 7.000                          |
| 30            | 68      | 21    | 5                | 8.000                         | 6.000                       | 11.000                      | 0.017                          | 6.485                          | 3.000                          | 10.000                         |
| 35            | 68      | 24    | 5                | 9.000                         | 7.000                       | 12.000                      | 0.003                          | 8.471                          | 5.000                          | 12.000                         |
| 40            | 68      | 28    | 8                | 13.000                        | 10.000                      | 16.000                      | 7.999e-04                      | 11.529                         | 8.000                          | 15.000                         |

The table reports five-percentage-point intervals for readability. The machine-readable source file retains every integer top-set share from 10% to 40%. Hypergeometric values are descriptive nonspatial comparators; inference uses within-sector permutations.

**Supplementary Table S10. Leave-one-coastal-sector-out calculations.**

| Terrain metric               | Omitted sector  | Omitted n | Remaining n | Spearman rho | Kendall tau | Two-sided p | Top n | Overlap |
|------------------------------|-----------------|-----------|-------------|--------------|-------------|-------------|-------|---------|
| connected_reference_land_pct | Atlantic-Celtic | 12        | 56          | -0.376       | -0.238      | 0.005       | 12    | 0       |

| Terrain metric               | Omitted sector                     | Omitted n | Remaining n | Spearman rho | Kendall tau | Two-sided p | Top n | Overlap |
|------------------------------|------------------------------------|-----------|-------------|--------------|-------------|-------------|-------|---------|
| connected_reference_land_pct | British-North-Sea                  | 19        | 49          | -0.359       | -0.221      | 0.014       | 10    | 0       |
| connected_reference_land_pct | Denmark-Norway                     | 2         | 66          | -0.371       | -0.239      | 0.002       | 14    | 1       |
| connected_reference_land_pct | Dutch-German-Bight                 | 4         | 64          | -0.327       | -0.202      | 0.009       | 13    | 0       |
| connected_reference_land_pct | eastern-Channel-southern-North-Sea | 10        | 58          | -0.482       | -0.320      | 2.000e-04   | 12    | 0       |
| connected_reference_land_pct | western-English-Channel            | 21        | 47          | -0.031       | -0.006      | 0.842       | 10    | 3       |
| below_reference_land_pct     | Atlantic-Celtic                    | 12        | 56          | -0.388       | -0.244      | 0.003       | 12    | 0       |
| below_reference_land_pct     | British-North-Sea                  | 19        | 49          | -0.352       | -0.207      | 0.014       | 10    | 0       |
| below_reference_land_pct     | Denmark-Norway                     | 2         | 66          | -0.344       | -0.213      | 0.006       | 14    | 1       |
| below_reference_land_pct     | Dutch-German-Bight                 | 4         | 64          | -0.305       | -0.179      | 0.016       | 13    | 0       |
| below_reference_land_pct     | eastern-Channel-southern-North-Sea | 10        | 58          | -0.426       | -0.273      | 0.001       | 12    | 0       |
| below_reference_land_pct     | western-English-Channel            | 21        | 47          | 0.015        | 0.025       | 0.920       | 10    | 3       |
| conditional_connectivity_pct | Atlantic-Celtic                    | 12        | 56          | 0.273        | 0.187       | 0.042       | 12    | 5       |
| conditional_connectivity_pct | British-North-Sea                  | 19        | 49          | 0.056        | 0.035       | 0.698       | 10    | 4       |
| conditional_connectivity_pct | Denmark-Norway                     | 2         | 66          | 0.101        | 0.065       | 0.421       | 14    | 5       |
| conditional_connectivity_pct | Dutch-German-Bight                 | 4         | 64          | 0.109        | 0.070       | 0.395       | 13    | 5       |
| conditional_connectivity_pct | eastern-Channel-southern-North-Sea | 10        | 58          | 0.113        | 0.072       | 0.397       | 12    | 4       |
| conditional_connectivity_pct | western-English-Channel            | 21        | 47          | -0.048       | -0.041      | 0.746       | 10    | 2       |
| connected_area_km2           | Atlantic-Celtic                    | 12        | 56          | -0.367       | -0.226      | 0.006       | 12    | 0       |
| connected_area_km2           | British-North-Sea                  | 19        | 49          | -0.343       | -0.196      | 0.016       | 10    | 0       |

| Terrain metric     | Omitted sector                     | Omitted n | Remaining n | Spearman rho | Kendall tau | Two-sided p | Top n | Overlap |
|--------------------|------------------------------------|-----------|-------------|--------------|-------------|-------------|-------|---------|
| connected_area_km2 | Denmark-Norway                     | 2         | 66          | -0.380       | -0.238      | 0.002       | 14    | 1       |
| connected_area_km2 | Dutch-German-Bight                 | 4         | 64          | -0.338       | -0.201      | 0.007       | 13    | 0       |
| connected_area_km2 | eastern-Channel-southern-North-Sea | 10        | 58          | -0.499       | -0.325      | 9.999e-05   | 12    | 0       |
| connected_area_km2 | western-English-Channel            | 21        | 47          | -0.043       | -0.005      | 0.771       | 10    | 3       |

**Supplementary Table S11. Sector-specific associations.**

| coastal sector                     | terrain metric               | terrain metric label     | n sites | spearman | kendall | status             |
|------------------------------------|------------------------------|--------------------------|---------|----------|---------|--------------------|
| Atlantic-Celtic                    | below_reference_land_pct     | lowland_prevalence       | 12      | -0.287   | -0.152  | estimated          |
| British-North-Sea                  | below_reference_land_pct     | lowland_prevalence       | 19      | -0.295   | -0.181  | estimated          |
| Denmark-Norway                     | below_reference_land_pct     | lowland_prevalence       | 2       | NA       | NA      | insufficient_n_lt4 |
| Dutch-German-Bight                 | below_reference_land_pct     | lowland_prevalence       | 4       | 0.800    | 0.667   | estimated          |
| eastern-Channel-southern-North-Sea | below_reference_land_pct     | lowland_prevalence       | 10      | -0.212   | -0.156  | estimated          |
| western-English-Channel            | below_reference_land_pct     | lowland_prevalence       | 21      | -0.743   | -0.505  | estimated          |
| Atlantic-Celtic                    | conditional_connectivity_pct | conditional_connectivity | 12      | -0.385   | -0.273  | estimated          |
| British-North-Sea                  | conditional_connectivity_pct | conditional_connectivity | 19      | 0.053    | -0.024  | estimated          |
| Denmark-Norway                     | conditional_connectivity_pct | conditional_connectivity | 2       | NA       | NA      | insufficient_n_lt4 |
| Dutch-German-Bight                 | conditional_connectivity_pct | conditional_connectivity | 4       | 0.400    | 0.333   | estimated          |
| eastern-Channel-southern-North-Sea | conditional_connectivity_pct | conditional_connectivity | 10      | 0.018    | 0.022   | estimated          |
| western-English-Channel            | conditional_connectivity_pct | conditional_connectivity | 21      | 0.322    | 0.216   | estimated          |
| Atlantic-Celtic                    | connected_reference_land_pct | connected_lowland_share  | 12      | -0.378   | -0.242  | estimated          |
| British-North-Sea                  | connected_reference_land_pct | connected_lowland_share  | 19      | -0.344   | -0.181  | estimated          |
| Denmark-Norway                     | connected_reference_land_pct | connected_lowland_share  | 2       | NA       | NA      | insufficient_n_lt4 |
| Dutch-German-Bight                 | connected_reference_land_pct | connected_lowland_share  | 4       | 0.400    | 0.333   | estimated          |
| eastern-Channel-southern-North-Sea | connected_reference_land_pct | connected_lowland_share  | 10      | -0.309   | -0.244  | estimated          |

| coastal sector                     | terrain metric                   | terrain metric label        | n sites | spearman | kendall | status             |
|------------------------------------|----------------------------------|-----------------------------|---------|----------|---------|--------------------|
| western-English-Channel            | connected_referenc<br>e_land_pct | connected_lowland<br>_share | 21      | -0.738   | -0.514  | estimated          |
| Atlantic-Celtic                    | connected_area_km<br>2           | connected_area              | 12      | -0.182   | -0.091  | estimated          |
| British-North-Sea                  | connected_area_km<br>2           | connected_area              | 19      | -0.377   | -0.193  | estimated          |
| Denmark-Norway                     | connected_area_km<br>2           | connected_area              | 2       | NA       | NA      | insufficient_n_lt4 |
| Dutch-German-Bight                 | connected_area_km<br>2           | connected_area              | 4       | 0.400    | 0.333   | estimated          |
| eastern-Channel-southern-North-Sea | connected_area_km<br>2           | connected_area              | 10      | -0.236   | -0.156  | estimated          |
| western-English-Channel            | connected_area_km<br>2           | connected_area              | 21      | -0.684   | -0.467  | estimated          |
| Atlantic-Celtic                    | connected_window<br>_pct         | connected_window<br>_share  | 12      | -0.182   | -0.091  | estimated          |
| British-North-Sea                  | connected_window<br>_pct         | connected_window<br>_share  | 19      | -0.377   | -0.193  | estimated          |
| Denmark-Norway                     | connected_window<br>_pct         | connected_window<br>_share  | 2       | NA       | NA      | insufficient_n_lt4 |
| Dutch-German-Bight                 | connected_window<br>_pct         | connected_window<br>_share  | 4       | 0.400    | 0.333   | estimated          |
| eastern-Channel-southern-North-Sea | connected_window<br>_pct         | connected_window<br>_share  | 10      | -0.236   | -0.156  | estimated          |
| western-English-Channel            | connected_window<br>_pct         | connected_window<br>_share  | 21      | -0.684   | -0.467  | estimated          |

**Supplementary Table S12. Moran spatial-autocorrelation diagnostics.**

| variable                     | distance<br>threshold km | moran i | expected i<br>under<br>randomization | permutation<br>two sided p | null q025 | null q975 | weight<br>definition                                                                                                | n permutations |
|------------------------------|--------------------------|---------|--------------------------------------|----------------------------|-----------|-----------|---------------------------------------------------------------------------------------------------------------------|----------------|
| coast_rp_rp10<br>_m          | 150.000                  | 0.302   | -0.015                               | 0.001                      | -0.175    | 0.161     | row-<br>standardized<br>binary great-<br>circle distance;<br>isolated<br>stations linked<br>to nearest<br>neighbour | 4999           |
| below_referenc<br>e_land_pct | 150.000                  | 0.758   | -0.015                               | 2.000e-04                  | -0.134    | 0.182     | row-<br>standardized<br>binary great-<br>circle distance;<br>isolated<br>stations linked<br>to nearest<br>neighbour | 4999           |

| variable                     | distance threshold km | moran i | expected i under randomization | permutation two sided p | null q025 | null q975 | weight definition                                                                            | n permutations |
|------------------------------|-----------------------|---------|--------------------------------|-------------------------|-----------|-----------|----------------------------------------------------------------------------------------------|----------------|
| conditional_connectivity_pct | 150.000               | -0.019  | -0.015                         | 0.959                   | -0.150    | 0.178     | row-standardized binary great-circle distance; isolated stations linked to nearest neighbour | 4999           |
| connected_reference_land_pct | 150.000               | 0.554   | -0.015                         | 0.001                   | -0.116    | 0.179     | row-standardized binary great-circle distance; isolated stations linked to nearest neighbour | 4999           |
| connected_area_km2           | 150.000               | 0.626   | -0.015                         | 2.000e-04               | -0.104    | 0.156     | row-standardized binary great-circle distance; isolated stations linked to nearest neighbour | 4999           |
| connected_window_pct         | 150.000               | 0.625   | -0.015                         | 2.000e-04               | -0.102    | 0.161     | row-standardized binary great-circle distance; isolated stations linked to nearest neighbour | 4999           |
| coast_ratio_ratio_m          | 250.000               | 0.110   | -0.015                         | 0.030                   | -0.112    | 0.106     | row-standardized binary great-circle distance; isolated stations linked to nearest neighbour | 4999           |
| below_reference_land_pct     | 250.000               | 0.543   | -0.015                         | 2.000e-04               | -0.094    | 0.109     | row-standardized binary great-circle distance; isolated stations linked to nearest neighbour | 4999           |

| variable                     | distance threshold km | moran i | expected i under randomization | permutation two sided p | null q025 | null q975 | weight definition                                                                            | n permutations |
|------------------------------|-----------------------|---------|--------------------------------|-------------------------|-----------|-----------|----------------------------------------------------------------------------------------------|----------------|
| conditional_connectivity_pct | 250.000               | 0.019   | -0.015                         | 0.480                   | -0.104    | 0.106     | row-standardized binary great-circle distance; isolated stations linked to nearest neighbour | 4999           |
| connected_reference_land_pct | 250.000               | 0.418   | -0.015                         | 6.000e-04               | -0.085    | 0.100     | row-standardized binary great-circle distance; isolated stations linked to nearest neighbour | 4999           |
| connected_area_km2           | 250.000               | 0.393   | -0.015                         | 2.000e-04               | -0.078    | 0.083     | row-standardized binary great-circle distance; isolated stations linked to nearest neighbour | 4999           |
| connected_window_pct         | 250.000               | 0.393   | -0.015                         | 2.000e-04               | -0.078    | 0.082     | row-standardized binary great-circle distance; isolated stations linked to nearest neighbour | 4999           |

**Supplementary Table S13. Maximum leave-one-station influence by terrain metric.**

| Terrain metric           | Most influential omitted station | Sector                             | Full rho | Leave-one-out rho | Change in rho |
|--------------------------|----------------------------------|------------------------------------|----------|-------------------|---------------|
| lowland_prevalence       | Hinkleypoint P003 Uk             | western-English-Channel            | -0.309   | -0.279            | 0.030         |
| conditional_connectivity | Hinkleypoint P003 Uk             | western-English-Channel            | 0.105    | 0.070             | -0.035        |
| connected_area           | Sheerness                        | eastern-Channel-southern-North-Sea | -0.341   | -0.372            | -0.031        |
| connected_lowland_share  | Sheerness                        | eastern-Channel-southern-North-Sea | -0.333   | -0.364            | -0.031        |
| connected_window_share   | Sheerness                        | eastern-Channel-southern-North-Sea | -0.341   | -0.372            | -0.031        |

**Supplementary Table S14. Four- versus eight-neighbour summary.**

| terrain metric               | n sites | rank spearman 4n vs 8n | top20 n | top20 overlap | membership changes | maximum eight minus four | median eight minus four |
|------------------------------|---------|------------------------|---------|---------------|--------------------|--------------------------|-------------------------|
| connected_reference_land_pct | 68      | 0.961                  | 14      | 11            | 6                  | 44.057                   | 0.007                   |

| terrain metric               | n sites | rank spearman<br>4n vs 8n | top20 n | top20 overlap | membership<br>changes | maximum eight<br>minus four | median eight<br>minus four |
|------------------------------|---------|---------------------------|---------|---------------|-----------------------|-----------------------------|----------------------------|
| connected_area_km2           | 68      | 0.967                     | 14      | 11            | 6                     | 23.550                      | 0.004                      |
| conditional_connectivity_pct | 68      | 0.901                     | 14      | 13            | 2                     | 91.527                      | 0.581                      |

**Supplementary Table S15. Stations whose top-20% membership changes between four and eight neighbours.**

| Station             | Terrain metric               | 4-neighbour value | 8-neighbour value | Membership change |
|---------------------|------------------------------|-------------------|-------------------|-------------------|
| Hoek van Holland    | connected_reference_land_pct | 2.640             | 46.697            | entered_under_8n  |
| Tregde 006 Norway   | connected_reference_land_pct | 5.707             | 5.768             | left_under_8n     |
| Lowestoft P024 Uk   | connected_reference_land_pct | 1.416             | 11.570            | entered_under_8n  |
| Immingham P026 Uk   | connected_reference_land_pct | 0.406             | 14.077            | entered_under_8n  |
| Holyhead P054 Uk    | connected_reference_land_pct | 5.703             | 6.109             | left_under_8n     |
| Barmouth P923 Uk    | connected_reference_land_pct | 5.886             | 6.323             | left_under_8n     |
| Hoek van Holland    | connected_area_km2           | 1.411             | 24.961            | entered_under_8n  |
| Tregde 006 Norway   | connected_area_km2           | 2.743             | 2.773             | left_under_8n     |
| Lowestoft P024 Uk   | connected_area_km2           | 0.670             | 5.470             | entered_under_8n  |
| Immingham P026 Uk   | connected_area_km2           | 0.249             | 8.659             | entered_under_8n  |
| Holyhead P054 Uk    | connected_area_km2           | 2.190             | 2.345             | left_under_8n     |
| Stmarys P231 Uk     | connected_area_km2           | 2.911             | 2.912             | left_under_8n     |
| Saint Gildas France | conditional_connectivity_pct | 99.308            | 99.308            | left_under_8n     |
| Heysham P050 Uk     | conditional_connectivity_pct | 99.103            | 99.402            | entered_under_8n  |

**Supplementary Table S16. Recorded Figure 3 case-selection rules and values.**

| Station            | Figure 3 role                       | Recorded<br>selection rule                                                                                                                   | RP10 (m) | Water rank | Terrain rank | Below-2 m<br>share (%) | Conditional<br>connection<br>(%) | Connected<br>share (%) | Connected<br>area (km2) |
|--------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|--------------|------------------------|----------------------------------|------------------------|-------------------------|
| Delfzijl Del<br>NI | stable<br>terrain-<br>priority case | Top-20%<br>connected-<br>land share,<br>local-DTM<br>IoU >= 0.75;<br>smallest<br>absolute<br>local-minus-<br>DeltaDTM<br>share<br>difference | 4.094    | 26         | 1            | 97.003                 | 99.692                           | 96.704                 | 61.303                  |

| Station           | Figure 3 role                        | Recorded selection rule                                                                                                         | RP10 (m) | Water rank | Terrain rank | Below-2 m share (%) | Conditional connection (%) | Connected share (%) | Connected area (km2) |
|-------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------|------------|--------------|---------------------|----------------------------|---------------------|----------------------|
| Newport Po57 Uk   | water-priority, lowland-limited case | Top-20% storm tide and conditional connectivity >= 90%; maximum terrain-minus-water rank displacement , then highest water rank | 8.602    | 3          | 67           | 0.164               | 90.528                     | 0.148               | 0.102                |
| Immingham Po26 Uk | connectivity-filtered case           | Below-2 m land share at or above the sample median; minimum conditional connectivity                                            | 4.528    | 22         | 49           | 20.142              | 2.014                      | 0.406               | 0.249                |

**Supplementary Table S17. Seven-site local-DTM product, datum and resampling cross-check.**

| Station   | Local terrain product                                           | Native datum                                    | Delta connected share (%) | Local connected share (%) | Delta conditional connection (%) | Local conditional connection (%) | Delta area (km2) | Local area (km2) | Connected-mask IoU | Elevation resampling                                                                          |
|-----------|-----------------------------------------------------------------|-------------------------------------------------|---------------------------|---------------------------|----------------------------------|----------------------------------|------------------|------------------|--------------------|-----------------------------------------------------------------------------------------------|
| Sheerness | Environment Agency LiDAR Composite DTM 2 m WCS, scaleFactor 0.2 | Ordnance Datum Newlyn / EA product native datum | 42.700                    | 9.630                     | 93.412                           | 33.626                           | 20.364           | 4.612            | 0.215              | provider WCS scaleFactor to nominal 10 m; interpolation kernel controlled by provider service |
| Newlyn    | Environment Agency LiDAR Composite DTM 2 m WCS, scaleFactor 0.2 | Ordnance Datum Newlyn / EA product native datum | 0.446                     | 0.417                     | 98.367                           | 98.931                           | 0.276            | 0.259            | 0.528              | provider WCS scaleFactor to nominal 10 m; interpolation kernel controlled by provider service |

| Station           | Local terrain product                                           | Native datum                                        | Delta connected share (%) | Local connected share (%) | Delta conditional connection (%) | Local conditional connection (%) | Delta area (km2) | Local area (km2) | Connected-mask IoU | Elevation resampling                                                                          |
|-------------------|-----------------------------------------------------------------|-----------------------------------------------------|---------------------------|---------------------------|----------------------------------|----------------------------------|------------------|------------------|--------------------|-----------------------------------------------------------------------------------------------|
| Lowestoft Po24 Uk | Environment Agency LiDAR Composite DTM 2 m WCS, scaleFactor 0.2 | Ordnance Datum Newlyn / EA product native datum     | 1.416                     | 6.165                     | 11.707                           | 46.234                           | 0.670            | 2.928            | 0.111              | provider WCS scaleFactor to nominal 10 m; interpolation kernel controlled by provider service |
| Immingham Po26 Uk | Environment Agency LiDAR Composite DTM 2 m WCS, scaleFactor 0.2 | Ordnance Datum Newlyn / EA product native datum     | 0.406                     | 0.079                     | 2.014                            | 0.819                            | 0.249            | 0.049            | 0.042              | provider WCS scaleFactor to nominal 10 m; interpolation kernel controlled by provider service |
| Denhelder Hel NL  | PDOK/Rijkswaterstaat AHN DTM 0.5 m WCS, scaleFactor 0.05        | Normaal Amsterdam s Peil / AHN product native datum | 71.730                    | 4.365                     | 98.327                           | 6.770                            | 20.855           | 1.274            | 0.055              | provider WCS scaleFactor to nominal 10 m; interpolation kernel controlled by provider service |
| Delfzijl Del NL   | PDOK/Rijkswaterstaat AHN DTM 0.5 m WCS, scaleFactor 0.05        | Normaal Amsterdam s Peil / AHN product native datum | 96.704                    | 89.784                    | 99.692                           | 99.513                           | 61.303           | 57.187           | 0.926              | provider WCS scaleFactor to nominal 10 m; interpolation kernel controlled by provider service |
| Hoek van Holland  | PDOK/Rijkswaterstaat AHN DTM 0.5 m WCS, scaleFactor 0.05        | Normaal Amsterdam s Peil / AHN product native datum | 2.640                     | 1.439                     | 5.484                            | 3.537                            | 1.411            | 0.772            | 0.217              | provider WCS scaleFactor to nominal 10 m; interpolation kernel controlled by provider service |

Supplementary Table S18. Excluded COAST-RP nearest-point matches.

| Station             | Latitude | Longitude | COAST-RP point                     | Distance (km) | Reason                                |
|---------------------|----------|-----------|------------------------------------|---------------|---------------------------------------|
| Cordemais France    | 47.277   | -1.890    | id_obs_donges_donges_france_refmar | 22.647        | excluded_nearest_coast_rp_point_gt6km |
| Le Pellerin France  | 47.205   | -1.765    | IBIS_GL_TS_MO_IFo00578             | 28.388        | excluded_nearest_coast_rp_point_gt6km |
| Nantes Usine Brulee | 47.204   | -1.574    | IBIS_GL_TS_MO_IFo00578             | 39.423        | excluded_nearest_coast_rp_point_gt6km |
| Donges France       | 47.306   | -2.088    | id_obs_donges_donges_france_refmar | 8.278         | excluded_nearest_coast_rp_point_gt6km |
| Montoir De Bretagne | 47.306   | -2.108    | id_obs_donges_donges_france_refmar | 6.897         | excluded_nearest_coast_rp_point_gt6km |
| Paimboeuf France    | 47.287   | -2.000    | id_obs_donges_donges_france_refmar | 14.337        | excluded_nearest_coast_rp_point_gt6km |

Supplementary Table S19. Focal GSSR and COAST-RP product-context diagnostics.

| Station              | GSSR empirical RP10 (m) | Bootstrap 2.5% | Bootstrap 97.5% | GSSR lower-bound RP10 | GSSR upper-bound RP10 | COAST-RP RP10 (m) | Within-5 km minimum | Within-5 km maximum | D10 cross-product difference (m) | Tidal range (m) | Tidal source status      |
|----------------------|-------------------------|----------------|-----------------|-----------------------|-----------------------|-------------------|---------------------|---------------------|----------------------------------|-----------------|--------------------------|
| Sheerness            | 1.363                   | 1.277          | 1.470           | 0.999                 | 1.705                 | 5.256             | 5.085               | 5.256               | 3.893                            | 5.200           | source_verified          |
| Newlyn               | 0.617                   | 0.563          | 0.673           | 0.469                 | 0.766                 | 3.346             | 3.346               | 3.348               | 2.729                            | 4.600           | source_verified          |
| Aberdeen             | 0.838                   | 0.789          | 0.882           | 0.663                 | 1.014                 | 2.822             | 2.816               | 2.822               | 1.984                            | 3.500           | source_verified          |
| Hoek van Holland     | 1.505                   | 1.430          | 1.622           | 1.204                 | 1.810                 | 3.373             | 3.360               | 3.373               | 1.868                            | 1.700           | indicative_context       |
| Brest                | 0.687                   | 0.645          | 0.847           | 0.517                 | 0.857                 | 4.313             | 4.233               | 4.321               | 3.626                            | 6.100           | indicative_context       |
| New York-The Battery | 1.295                   | 1.067          | 1.459           | 1.064                 | 1.525                 | 1.311             | 1.311               | 1.311               | 0.016                            | 1.380           | source_verified          |
| Hong Kong            | 1.026                   | 0.751          | 1.143           | 0.829                 | 1.225                 | 1.823             | 1.823               | 1.823               | 0.797                            | 1.900           | official_general_context |
| Charleston           | 0.843                   | 0.731          | 1.259           | 0.630                 | 1.069                 | 1.346             | 1.346               | 1.356               | 0.503                            | 1.591           | source_verified          |

Supplementary Table S20. Source-verified tidal stations plotted in Supplementary Figure S1.

| Station  | Spring tidal range (m) | COAST-RP RP10 (m) | Connected land share (%) | Source                                                                     | URL                                                                                                                   |
|----------|------------------------|-------------------|--------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Newlyn   | 4.600                  | 3.346             | 0.446                    | National Tidal and Sea Level Facility highest/lowest predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> |
| Aberdeen | 3.500                  | 2.822             | 0.982                    | National Tidal and Sea Level Facility highest/lowest predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> |

| Station              | Spring tidal range (m) | COAST-RP RP10 (m) | Connected land share (%) | Source                                                                           | URL                                                                                                                   |
|----------------------|------------------------|-------------------|--------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Sheerness            | 5.200                  | 5.256             | 42.700                   | National Tidal and Sea Level Facility<br>highest/lowest<br>predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> |
| Lerwick Po41 Uk      | 1.600                  | 1.508             | 2.338                    | National Tidal and Sea Level Facility<br>highest/lowest<br>predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> |
| Portpatrick Po63 Uk  | 3.400                  | 2.887             | 0.546                    | National Tidal and Sea Level Facility<br>highest/lowest<br>predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> |
| Fishguard Po55 Uk    | 3.700                  | 2.998             | 0.178                    | National Tidal and Sea Level Facility<br>highest/lowest<br>predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> |
| Milfordhaven Po56 Uk | 6.100                  | 4.376             | 0.472                    | National Tidal and Sea Level Facility<br>highest/lowest<br>predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> |
| Lowestoft Po24 Uk    | 1.800                  | 2.379             | 1.416                    | National Tidal and Sea Level Facility<br>highest/lowest<br>predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> |
| Wick Po35 Uk         | 2.700                  | 2.560             | 1.552                    | National Tidal and Sea Level Facility<br>highest/lowest<br>predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> |
| Stornoway Po42 Uk    | 3.900                  | 2.980             | 1.911                    | National Tidal and Sea Level Facility<br>highest/lowest<br>predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> |
| Immingham Po26 Uk    | 6.000                  | 4.528             | 0.406                    | National Tidal and Sea Level Facility<br>highest/lowest<br>predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> |
| Millport Po49 Uk     | 2.800                  | 2.717             | 1.537                    | National Tidal and Sea Level Facility<br>highest/lowest<br>predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> |
| Holyhead Po54 Uk     | 4.800                  | 3.315             | 5.703                    | National Tidal and Sea Level Facility<br>highest/lowest<br>predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> |

**Supplementary Table S21. Complete tidal metadata source audit.**

| Station | Metric                          | Range (m) | Source status   | Source                                                                           | URL                                                                                                                   | Notes                                                                               |
|---------|---------------------------------|-----------|-----------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Newlyn  | mean spring range = MHWS - MLWS | 4.600     | source_verified | National Tidal and Sea Level Facility<br>highest/lowest<br>predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> | Values parsed from the NTSLF AJAX table; heights are given relative to chart datum. |

| Station                | Metric                          | Range (m) | Source status                | Source                                                                                                                    | URL                                                                                                                                                                                               | Notes                                                                                      |
|------------------------|---------------------------------|-----------|------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Hoek van Holland       | spring range context            | 1.700     | indicative_context           | Port of Rotterdam/Rijkswaterstaat candidate source identified; exact station spring range not yet extracted automatically | <a href="https://www.portofrotterdam.com/nl/actuele-informatie/weer-getijde-en-waterdiepte">https://www.portofrotterdam.com/nl/actuele-informatie/weer-getijde-en-waterdiepte</a>                 | Retain as context until HydroMeteoBundel or Rijkswaterstaat station statistics are parsed. |
| Brest                  | spring range context            | 6.100     | indicative_context           | SHOM/REFMAR tidal-prediction definitions identified; exact station spring range not yet extracted automatically           | <a href="https://refmar.shom.fr/sites/default/files/2025-01/GT-TSH_CatD_Fiche_Prediction_maree.pdf">https://refmar.shom.fr/sites/default/files/2025-01/GT-TSH_CatD_Fiche_Prediction_maree.pdf</a> | Retain as context until SHOM station PMVE/BMVE or harmonic metadata are parsed.            |
| Aberdeen               | mean spring range = MHWS - MLWS | 3.500     | source_verified              | National Tidal and Sea Level Facility highest/lowest predicted tides table                                                | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a>                                                                             | Values parsed from the NTSLF AJAX table; heights are given relative to chart datum.        |
| Sheerness              | mean spring range = MHWS - MLWS | 5.200     | source_verified              | National Tidal and Sea Level Facility highest/lowest predicted tides table                                                | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a>                                                                             | Values parsed from the NTSLF AJAX table; heights are given relative to chart datum.        |
| Esbjerg 130121 Denmark | not available                   | NA        | missing_or_not_yet_extracted | NA                                                                                                                        | NA                                                                                                                                                                                                | Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.        |
| Denhelder Hel NL       | not available                   | NA        | missing_or_not_yet_extracted | NA                                                                                                                        | NA                                                                                                                                                                                                | Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.        |
| Lerwick PO41 UK        | mean spring range = MHWS - MLWS | 1.600     | source_verified              | National Tidal and Sea Level Facility highest/lowest predicted tides table                                                | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a>                                                                             | Values parsed from the NTSLF AJAX table; heights are given relative to chart datum.        |
| Cuxhaven Germany       | not available                   | NA        | missing_or_not_yet_extracted | NA                                                                                                                        | NA                                                                                                                                                                                                | Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.        |
| Delfzijl Del NL        | not available                   | NA        | missing_or_not_yet_extracted | NA                                                                                                                        | NA                                                                                                                                                                                                | Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.        |

| Station              | Metric                          | Range (m) | Source status                | Source                                                                     | URL                                                                                                                   | Notes                                                                               |
|----------------------|---------------------------------|-----------|------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Portpatrick Po63 Uk  | mean spring range = MHWS - MLWS | 3.400     | source_verified              | National Tidal and Sea Level Facility highest/lowest predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> | Values parsed from the NTSLF AJAX table; heights are given relative to chart datum. |
| Fishguard Po55 Uk    | mean spring range = MHWS - MLWS | 3.700     | source_verified              | National Tidal and Sea Level Facility highest/lowest predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> | Values parsed from the NTSLF AJAX table; heights are given relative to chart datum. |
| Milfordhaven Po56 Uk | mean spring range = MHWS - MLWS | 6.100     | source_verified              | National Tidal and Sea Level Facility highest/lowest predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> | Values parsed from the NTSLF AJAX table; heights are given relative to chart datum. |
| Tregde 006 Norway    | not available                   | NA        | missing_or_not_yet_extracted | NA                                                                         | NA                                                                                                                    | Non-UK/non-US extended-site tidal metadata not automatically extracted in this run. |
| Lowestoft Po24 Uk    | mean spring range = MHWS - MLWS | 1.800     | source_verified              | National Tidal and Sea Level Facility highest/lowest predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> | Values parsed from the NTSLF AJAX table; heights are given relative to chart datum. |
| Wick Po35 Uk         | mean spring range = MHWS - MLWS | 2.700     | source_verified              | National Tidal and Sea Level Facility highest/lowest predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> | Values parsed from the NTSLF AJAX table; heights are given relative to chart datum. |
| Stornoway Po42 Uk    | mean spring range = MHWS - MLWS | 3.900     | source_verified              | National Tidal and Sea Level Facility highest/lowest predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> | Values parsed from the NTSLF AJAX table; heights are given relative to chart datum. |
| Le Conquet France    | not available                   | NA        | missing_or_not_yet_extracted | NA                                                                         | NA                                                                                                                    | Non-UK/non-US extended-site tidal metadata not automatically extracted in this run. |
| Port Tudy France     | not available                   | NA        | missing_or_not_yet_extracted | NA                                                                         | NA                                                                                                                    | Non-UK/non-US extended-site tidal metadata not automatically extracted in this run. |
| Cherbourg France     | not available                   | NA        | missing_or_not_yet_extracted | NA                                                                         | NA                                                                                                                    | Non-UK/non-US extended-site tidal metadata not automatically extracted in this run. |

| Station              | Metric                          | Range (m) | Source status                | Source                                                                     | URL                                                                                                                   | Notes                                                                               |
|----------------------|---------------------------------|-----------|------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Saint Nazaire France | not available                   | NA        | missing_or_not_yet_extracted | NA                                                                         | NA                                                                                                                    | Non-UK/non-US extended-site tidal metadata not automatically extracted in this run. |
| Saint Gildas France  | not available                   | NA        | missing_or_not_yet_extracted | NA                                                                         | NA                                                                                                                    | Non-UK/non-US extended-site tidal metadata not automatically extracted in this run. |
| Le Havre France      | not available                   | NA        | missing_or_not_yet_extracted | NA                                                                         | NA                                                                                                                    | Non-UK/non-US extended-site tidal metadata not automatically extracted in this run. |
| Dunkerque France     | not available                   | NA        | missing_or_not_yet_extracted | NA                                                                         | NA                                                                                                                    | Non-UK/non-US extended-site tidal metadata not automatically extracted in this run. |
| Roscoff France       | not available                   | NA        | missing_or_not_yet_extracted | NA                                                                         | NA                                                                                                                    | Non-UK/non-US extended-site tidal metadata not automatically extracted in this run. |
| Boulogne Sur Mer     | not available                   | NA        | missing_or_not_yet_extracted | NA                                                                         | NA                                                                                                                    | Non-UK/non-US extended-site tidal metadata not automatically extracted in this run. |
| Dieppe France        | not available                   | NA        | missing_or_not_yet_extracted | NA                                                                         | NA                                                                                                                    | Non-UK/non-US extended-site tidal metadata not automatically extracted in this run. |
| Immingham P026 Uk    | mean spring range = MHWS - MLWS | 6.000     | source_verified              | National Tidal and Sea Level Facility highest/lowest predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> | Values parsed from the NTSLF AJAX table; heights are given relative to chart datum. |
| Millport P049 Uk     | mean spring range = MHWS - MLWS | 2.800     | source_verified              | National Tidal and Sea Level Facility highest/lowest predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> | Values parsed from the NTSLF AJAX table; heights are given relative to chart datum. |

| Station          | Metric                          | Range (m) | Source status   | Source                                                                     | URL                                                                                                                   | Notes                                                                               |
|------------------|---------------------------------|-----------|-----------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Holyhead PO54 Uk | mean spring range = MHWS - MLWS | 4.800     | source_verified | National Tidal and Sea Level Facility highest/lowest predicted tides table | <a href="https://ntslf.org/tides/uk-network/predictions-hilo">https://ntslf.org/tides/uk-network/predictions-hilo</a> | Values parsed from the NTSLF AJAX table; heights are given relative to chart datum. |

## Supplementary Table S22. GSSR metadata-field and filter audit.

| audit item                       | value                                         | interpretation                                                                                                                                |
|----------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| available_metadata_fields        | FIELD1;corr_n;num_95pcs;num_year;path;rmse;tg | corr_n is retained as the source metadata reconstruction correlation; no separate extreme-event skill field is present in the local metadata. |
| eligible_corr_geo0.45_years_ge20 | 62                                            | Supporting GSSR-quality subset count only; this filter does not define the primary COAST-RP-terrain sample.                                   |
| eligible_corr_geo0.45_years_ge25 | 47                                            | Supporting GSSR-quality subset count only; this filter does not define the primary COAST-RP-terrain sample.                                   |
| eligible_corr_geo0.45_years_ge30 | 38                                            | Supporting GSSR-quality subset count only; this filter does not define the primary COAST-RP-terrain sample.                                   |
| eligible_corr_geo0.55_years_ge20 | 61                                            | Supporting GSSR-quality subset count only; this filter does not define the primary COAST-RP-terrain sample.                                   |
| eligible_corr_geo0.55_years_ge25 | 46                                            | Supporting GSSR-quality subset count only; this filter does not define the primary COAST-RP-terrain sample.                                   |
| eligible_corr_geo0.55_years_ge30 | 38                                            | Supporting GSSR-quality subset count only; this filter does not define the primary COAST-RP-terrain sample.                                   |
| eligible_corr_geo0.65_years_ge20 | 59                                            | Supporting GSSR-quality subset count only; this filter does not define the primary COAST-RP-terrain sample.                                   |
| eligible_corr_geo0.65_years_ge25 | 45                                            | Supporting GSSR-quality subset count only; this filter does not define the primary COAST-RP-terrain sample.                                   |
| eligible_corr_geo0.65_years_ge30 | 38                                            | Supporting GSSR-quality subset count only; this filter does not define the primary COAST-RP-terrain sample.                                   |

## Supplementary Data provenance

Machine-readable source files for all tables are retained under `data/figure_source/`. The scientific-integrity review records current hashes, invariants and known external-deposit placeholders. References are listed in the main manuscript.